SMART STUDY PLANNER – PROJECT REPORT

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INTRODUCTION

Managing time is a problem for students especially when they are getting ready for exams. I know this from my experience as a first-year engineering student. I had a time figuring out how much time to spend on each subject. I would spend time on the subjects I was good at and avoid the hard ones. This meant I was not preparing well for all my subjects.

When exams were coming up I would get confused about where to start and which subject I needed to focus on. This would cause me a lot of stress at the minute and I would not use my time well.

To solve this problem I decided to make a Smart Study Planner. This is a system that uses intelligence to help students make a good study schedule. The main idea is to help students plan their study time based on how hard each subject's when their exams are.

The system is simple to use. You just need to tell it the names of your subjects how hard they're when your exams are and how many hours you have to study. Then

it makes a 7-day study plan that helps you use your time wisely.

The system does not just assign time randomly. It uses some artificial intelligence ideas like being greedy and making decisions based on rules of thumb. This means that the important subjects get more time but you still have a balanced schedule.

My goal for this project is to make something for students. I also want to see how simple artificial intelligence techniques can be used to solve problems in a practical way. I am using the Smart Study Planner to help students like me manage their time better and do well in their exams. The Smart Study Planner is a tool that can really make a difference, for students.

1. Problem Identification

- In my life as a student I have seen that students often struggle to manage their time for studying effectively.
- Most students:
  - Focus too much on the easy subjects
  - Ignore the difficult subjects
  - Start preparing for exams too late
- This leads to:
  - Poor performance
  - A lot of stress
  - Inefficient use of time
- The Smart Study Planner is designed to solve this problem.
- The problem becomes more serious when:
  - There are multiple subjects
  - Different levels of difficulty exist
  - Different exam dates make planning harder
- This is where the Smart Study Planner comes in to help students manage their time effectively.

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2. Why This Problem Matters

- Managing time effectively is very important for doing well in school.

- Without a plan:
  - Students waste time on less important subjects
  - They do not focus on high-priority subjects
- This makes preparation unbalanced.
- The Smart Study Planner helps students to solve this problem.
- Benefits:
  - Study efficiently
  - Reduce stress before exams
  - Improve overall performance
- This makes the problem very relevant and practical in student life.
- The Smart Study Planner is designed to help students achieve their goals.

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3. Approach to the Solution

- Designed the Smart Study Planner as an AI-based system.
- It generates an optimized study plan.
- Takes inputs such as:
  - Name of the subject
  - Level of difficulty
  - Date of the exam
  - Number of study hours per day
- Based on inputs:
  - Calculates a priority score for each subject
  - Distributes study time accordingly

- The Smart Study Planner helps students use their time efficiently.

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4. Key Design Decisions

4.1 Use of Greedy Algorithm

- Chose this approach because:
  - It allocates time to the most important subject first
- Advantages:
  - Efficient
  - Simple to implement
- Suitable for real-time decision-making problems.

4.2 Heuristic Function Design

- Designed a function to determine subject importance:
  - Priority = (Difficulty × 0.7) + (Urgency × 0.3)
  - Urgency = 1 / Days Left
- Ensures:
  - Difficult subjects get more weight
  - Near exams get higher urgency

4.3 Proportional Time Allocation

- Used proportional allocation instead of fixed hours:
  - Time = (Subject Priority / Total Priority) × Total Study Hours
- Ensures:

    - Fair distribution of time
    - High-priority subjects get more time

4.4 Dynamic Scheduling

- Added slight variation each day
- Generated explanations such as:
    - "Exam near"
    - "High difficulty"
- Makes the planner behave like an intelligent assistant.

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5. Implementation Details

- **Frontend:**
    - HTML
    - CSS
    - JavaScript
- Features:
    - User input interface
    - Display of schedule
    - Chart.js for graphs
    - jsPDF for PDF export
- **Backend:**
    - Python with Flask
    - REST API for schedule generation
    - Heuristic-based logic implementation

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6. Challenges Faced

6.1 Designing the Heuristic

- Difficult to balance:
  - Difficulty
  - Urgency
- Tested multiple weight combinations before finalizing.

6.2 Frontend-Backend Integration

- Faced connection issues due to CORS
- Solved using Flask-CORS library

6.3 Making Output Realistic

- Early versions produced rigid schedules
- Added:
  - Variation
  - Explanations
- Improved realism significantly

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7. Results and Observations

- The Smart Study Planner:
  - Correctly prioritizes subjects
  - Distributes study time efficiently
  - Generates a 7-day schedule
  - Provides visual feedback using graphs
- Output:
  - Feels logical
  - Aligns with human planning

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8. What I Learned

- Learned how AI concepts like:
  - Algorithms
  - Heuristics
    can solve real-world problems
- Gained knowledge in:
  - System design
  - Frontend-backend integration
  - User experience and visualization

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9. Future Improvements

- Add learning based on user performance
- Integrate with calendar apps
- Develop a mobile application
- Use machine learning for better predictions

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10. Conclusion

- The Smart Study Planner demonstrates how simple AI techniques solve real-world problems
- Combines:
  - Algorithms
  - Heuristic functions
- Provides an efficient and practical study planning solution

- Highlights importance of applying AI to real problems rather than just theory
- The Smart Study Planner is a strong example of this